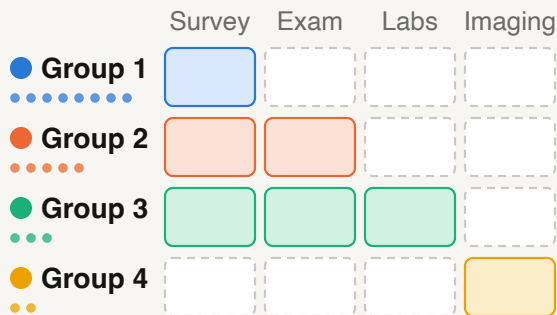


(a) Patients carry different tests



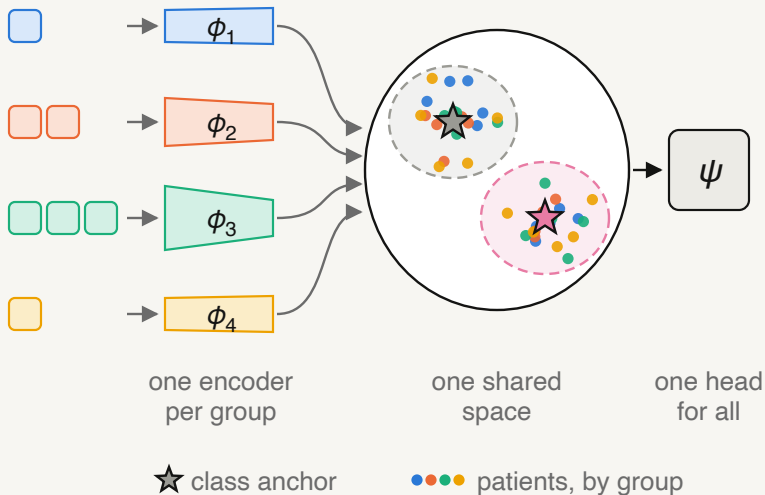
Group 4 shares no test with the others

■ measured ■ never ordered ● patients

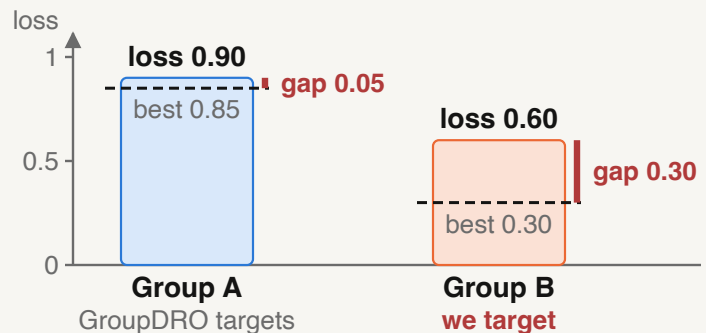
(b) How to share: one model for everyone

	keeps all data	imputes nothing	groups share	no overlap needed
Shared columns only	✗	✓	✓	✗
Imputation	✓	✗	✓	✓
One model per group	✓	✓	✗	✗
Modality fusion (MoE)	✓	✓	✓	✗
Ours	✓	✓	✓	✓

(c) Ours: share in a latent space



(d) What to optimise: the largest gap



	robust to group mix	protects worst group	fair to hard groups
ERM	✗	✗	✗
GroupDRO	✓	✓	✗
Ours: regret	✓	✓	✓